Section 1: Introduction

Value-at-Risk (VaR) remains one of the most widely used tools in market-risk measurement because it converts the lower tail of a portfolio return distribution into a single capital-relevant loss threshold. For a regulator, a portfolio manager, or an internal risk-control desk, the practical object is not the unconditional variance of returns, but the probability that the next trading day will produce a loss beyond a specified quantile. The quality of a VaR model therefore matters directly for capital allocation, risk limits, and the interpretation of stress periods. This forecasting problem is difficult for daily equity returns because the empirical distribution is not close to Gaussian and the conditional scale of returns changes sharply over time. In the SPY data used in this paper, daily log returns display heavy tails, volatility clustering, and clear transitions between calm and crisis regimes. These features imply that a simple normal VaR benchmark can be statistically convenient but economically misleading: it can understate the probability of extreme losses precisely when tail-risk measurement is most important.

The empirical challenge is not only to estimate a low quantile, but to estimate a conditional low quantile. A model may have approximately correct average coverage over the whole sample while still producing clustered violations during high-volatility periods. Conversely, a model may react well to volatility regimes while still producing too many violations in total. This distinction is central to statistical VaR backtesting. Unconditional coverage tests, such as the Kupiec test, ask whether the number of violations is consistent with the nominal tail probability. Independence and conditional-coverage tests, following the logic of Christoffersen's framework, ask whether violations arrive in a dynamically acceptable way. A credible empirical VaR model must therefore address both the marginal level of the lower tail and the time-series dependence in that tail.

The existing literature provides three major modelling routes, each with a different strength and a different limitation. The first route is non-parametric Historical Simulation. Its appeal is transparency: VaR is estimated directly from past returns and does not require a parametric distributional assumption. Hybrid or age-weighted variants, such as Boudoukh, Richardson and Whitelaw (1998), improve ordinary Historical Simulation by giving greater influence to recent observations. This is useful when volatility regimes change. At the same time, Historical Simulation remains backward-looking and can be slow to adapt when the rolling window contains observations from a different market state. Pritsker (2006) emphasizes that historical windows may also give a false sense of tail precision when extreme events are sparse. Thus, Historical Simulation preserves empirical tail information, but it does not by itself provide a structural model for conditional volatility dynamics.

The second route is the conditional heteroskedasticity literature. Engle's (1982) ARCH model and Bollerslev's (1986) GARCH model directly target the volatility clustering that is characteristic of financial returns. Student-t innovations add a heavy-tailed standardized shock distribution, and asymmetric extensions such as GJR-GARCH allow negative shocks to have a different volatility effect from positive shocks. These models are statistically natural for daily equity returns because they separate a time-varying volatility scale from an innovation distribution. They can improve the timing of VaR violations by raising predicted risk after large shocks. However, the same parametric structure can also be too rigid. A GARCH-t specification may reduce violation clustering while still failing unconditional coverage if the estimated tail level is too shallow. Realized-measure extensions, using variables such as realized variance and bipower variation, can add information about latent volatility, but a linear variance-side recursion is still not guaranteed to correct VaR calibration.

The third route is direct conditional quantile modelling. Quantile regression, introduced by Koenker and Bassett (1978), gives a natural statistical language for VaR because VaR is itself a conditional quantile. Neural-network quantile regression extends this idea by allowing nonlinear interactions among lagged returns, volatility proxies, and other state variables. This flexibility is attractive because the lower tail of equity returns may depend on market conditions in ways that are not well captured by a small parametric variance equation. The weakness is finite-sample learning. In a 1000-day rolling window, the 1% tail contains only about 10 observations. A neural network trained directly on such windows is asked to learn both the conditional volatility state and the absolute lower-tail level from very few extreme events. Without additional structure, flexibility can therefore increase estimation variance and produce unstable VaR forecasts.

This paper studies one-day-ahead VaR forecasting for SPY daily log returns under a unified rolling-window backtesting framework. The empirical comparison covers three model families. Section 3 evaluates ordinary Historical Simulation, time-weighted Historical Simulation, and KDE-smoothed time-weighted Historical Simulation. These models test how far empirical-tail information and recency weighting can go without an explicit conditional variance equation. Section 4 estimates GARCH-type conditional volatility models with Student-t innovations, including GARCH(1,1)-t, GJR-GARCH-t, and realized-measure GARCH-X robustness specifications using rv5 and bv. These models test the value of parametric volatility dynamics, leverage effects, and realized volatility information. Section 5 evaluates neural quantile regression models, including direct MLP quantile regression and a GARCH-anchored neural correction. All models are assessed using the same rolling forecast timing and the same backtesting criteria, so the comparison is not driven by inconsistent evaluation rules.

The central argument of the paper is that accurate VaR forecasting in this setting requires three components at the same time: heavy-tail information, conditional volatility dynamics, and controlled nonlinear flexibility. Historical Simulation contributes the first component but is weak in dynamic adaptation. GARCH-type models contribute the second component but can remain biased in unconditional coverage. Direct neural quantile regression contributes the third component in principle, but it is too unconstrained for rare-event estimation in small rolling windows. The paper therefore uses the failures and strengths of the individual model classes to motivate a hybrid specification rather than treating the neural network as a standalone replacement for classical risk models.

The main methodological contribution is the conservative GARCH-anchored neural quantile correction, referred to as Model C2. The model begins with a GARCH-t VaR forecast as a structured baseline. A neural network then learns a nonlinear correction using lagged return features, realized volatility proxies, GARCH VaR forecasts, and the GARCH volatility forecast. The key design choice is the direction of the correction. The GARCH-t backtests show that the baseline improves violation timing but leaves the VaR level too shallow on average. Model C2 converts this empirical diagnostic into an architectural restriction by using a softplus correction that can only move the baseline VaR in a more conservative direction. This is not an arbitrary penalty. It is a statistical bias-variance tradeoff: once the direction of the GARCH calibration bias is known, the network does not need to learn both the direction and the magnitude of the adjustment from a small number of tail observations.

This design also gives the neural model a clearer interpretation. The neural network is not responsible for discovering the entire conditional risk structure from scratch. The GARCH component supplies a volatility and heavy-tail anchor, while the neural component adjusts the VaR level conditional on additional state information. In this sense, the hybrid model is closer to a structured statistical correction than to an unrestricted machine-learning forecast. The ARCH-LM residual evidence and the GARCH-family backtests provide the motivation for retaining volatility structure, while the realized-measure variables provide information that may enter the tail risk function nonlinearly. The softplus restriction then prevents the correction from adding avoidable instability in the extreme tail.

The empirical findings support this integrated interpretation. Historical Simulation methods confirm that empirical tail information is useful, especially when recent observations receive greater weight and the empirical quantile is smoothed. However, they remain vulnerable to regime changes and violation clustering. GARCH-type models substantially improve the independence of VaR violations, but the symmetric GARCH-t benchmark still produces too many violations in total. Direct neural quantile regression performs poorly in the extreme tail, which is consistent with the difficulty of learning rare 1% events from rolling windows. The conservative GARCH-anchored model gives the most balanced result. Model C2 records failure rates of approximately 0.95%, 4.96%, and 8.90% at the 1%, 5%, and 10% VaR levels, with Kupiec p-values of 0.7823, 0.9288, and 0.0557. These results are much closer to nominal unconditional coverage than the direct neural models and the GARCH-family baselines.

The conclusion is deliberately qualified. Model C2 is the most defensible final specification in this project, but it does not completely solve dynamic extreme-tail forecasting. Its 5% and 10% independence diagnostics are acceptable, while the 1% Christoffersen p-value remains weak. This means that the most extreme violations still contain residual clustering. The evidence should therefore not be read as a general claim that neural networks dominate classical VaR methods. The stronger and more statistically defensible claim is that neural flexibility becomes useful when it is anchored to a meaningful conditional-risk structure and constrained in a direction supported by prior diagnostics.

The remainder of the paper proceeds as follows. Section 2 presents the data and common backtesting framework; Sections 3-5 evaluate Historical Simulation, GARCH-family models, and neural quantile methods; Section 6 integrates the evidence and concludes.